

Data acquisition

5-minute WIN futures
N = 15,057 bars

Feature representation

Technical indicators
Information Gain: 7 features

Chronological protocol

60% train | 20% validation | 20% test
No shuffling

Equal-budget search

1,500 evaluations per seed
RS, GA, PSO, DE, GWO

Validation selection

Candidate ranking on validation data
0.60 MCC + 0.40 F1

Locked evaluation

Held-out test: MCC and accuracy
Statistical and financial assessment